Chapter-04 + Chapter-04 Exercises (This Color means previous year questions)

❖ **Write the characteristics of GSM and its disadvantages. What has changed, what is the same if you compare GSM with today's systems?**

Characteristics:

- Communication: mobile, wireless communication; support for voice and data services
- Total mobility: international access, chip-card enables use of access points of different providers
- Worldwide connectivity: one number, the network handles localization
- High capacity: better frequency efficiency, smaller cells, more customers per cell
- High transmission quality: high audio quality and reliability for wireless, uninterrupted phone calls at higher speeds (e.g., from cars, trains)
- Security functions: access control, authentication via chip-card and PIN

Disadvantages:

- There is no perfect system!!
- no end-to-end encryption of user data
- no full ISDN bandwidth of 64 kbit/s to the user, no transparent B-channel
- reduced concentration while driving
- electromagnetic radiation
- abuse of private data possible

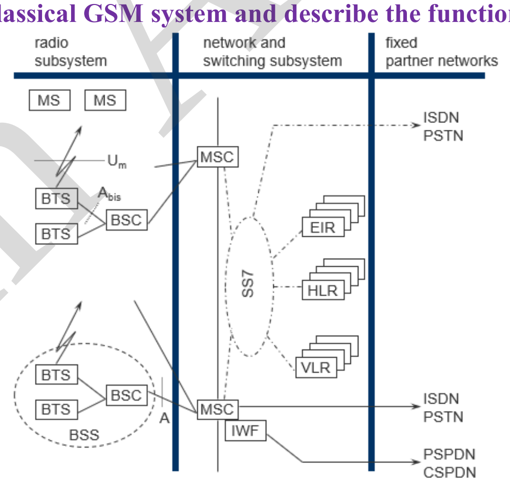
What has changed:

- Data speed: GSM (up to 50 kbps) → Today's systems (4G: 100+ Mbps, 5G: Gbps).
- Coverage & capacity: Modern systems support higher user density and global coverage.
- Services: GSM mainly voice & SMS → Today's systems support video calls, streaming, IoT, and high-speed internet.

What is the same:

- Core goal: mobile communication for voice and data.
- Cellular network structure with base stations and mobile switching centers.
- Subscriber identification using SIM cards (evolved but still used).

❖ **Draw the architecture of the classical GSM system and describe the functionality of the components.**



The GSM system consists of several main components that work together to provide mobile communication. The **Mobile Station (MS)** refers to the subscriber's mobile phone or device used to access the network. The **Base Station Subsystem (BSS)** includes the **Base Transceiver Station (BTS)**, commonly known as the cell tower, and the **Base Station Controller (BSC)**, which manages multiple BTSs and coordinates their functions. The **Network and Switching Subsystem (NSS)** forms the core of the GSM network, with key elements such as the **Mobile Switching Center (MSC)**, which handles call switching and connections to other networks; the **Home Location Register (HLR)**, a central database storing permanent subscriber information; and the **Visitor Location Register (VLR)**, a temporary database that maintains information about subscribers currently in its coverage area. Additionally, the GSM network interfaces with **Fixed Partner Networks**, including the **Public Switched Telephone Network (PSTN)** and the **Integrated Services Digital Network (ISDN)**, allowing seamless communication with external networks.

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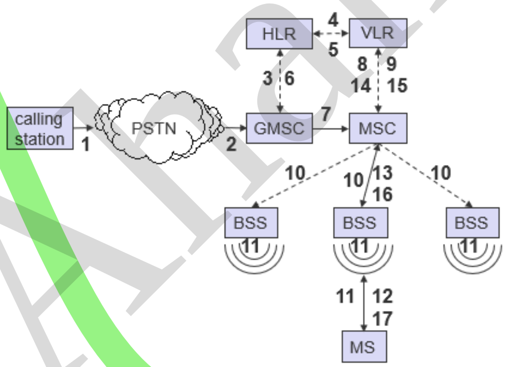
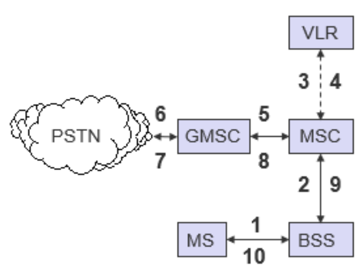
- ❖ **Why does GSM separate the MS and SIM?**
  SIM stores subscriber identity securely; MS handles device hardware. Allows portability across devices.
- ❖ **What is the advantage/disadvantage of an eSIM?**
  - Advantage: Remote provisioning, no physical SIM swap needed.
  - Disadvantage: Harder to remove/replace manually, depends on carrier support.
- ❖ **Go once more through the channel/slot structure and understand where GSM applies what type of multiplexing!**
  - FDM (Frequency Division Multiplexing): divides spectrum into 200 kHz carrier channels.
  - TDMA (Time Division Multiple Access): each carrier is divided into 8 time slots, each serving one user per frame.
- ❖ **How does GSM adapt to varying distances between MS and BTS? What is the role of the guard spaces?**
  - Timing Advance (TA): adjusts MS transmission to align with BTS timing.
  - Guard spaces: prevent time slot overlap caused by propagation delay.
- ❖ **What limits the radius of a GSM cell?**
  - Maximum ~35 km due to TDMA timing limits.
  - Also limited by signal strength, interference, and propagation delay.
- ❖ **What is the maximum data rate per time slot? Why is the real data rate offered that much lower?**
  - Maximum: 22.8 kbps per slot.
  - Real rate: ~9.6–14.4 kbps due to error correction, signaling, and channel coding overhead.
- ❖ **Draw the flowchart of Mobile Terminated Call (MTC).**

1: calling a GSM subscriber
2: forwarding call to GMSC
3: signal call setup to HLR
4, 5: request MSRN from VLR
6: forward responsible MSC to GMSC
7: forward call to current MSC
8, 9: get current status of MS
10, 11: paging of MS
12, 13: MS answers
14, 15: security checks
16, 17: set up connection



- ❖ **Draw the flowchart of Mobile Originated Call (MOC).**

1, 2: connection request
3, 4: security check
5-8: check resources (free circuit)
9-10: set up call



- ❖ **How does the GSM system locate an MS? How precise is the localization?**

  - ➢ GSM locates a mobile using the cell ID of the serving base station.
  - ➢ Precision can be improved with Timing Advance (TA) or triangulation using multiple BTS.
  - ➢ Accuracy: typically 100–500 meters in urban areas, up to a few kilometers in rural areas.

- **What are the key mechanisms and algorithms used for data encryption in GSM, and how do they contribute to ensuring the security and privacy of communications in a GSM network?**

Key Mechanisms / Algorithms:

- A3 algorithm: Used for authentication of the subscriber. Confirms that the MS is valid on the network.
- A8 algorithm: Generates session keys (Kc) from the subscriber key (Ki) for encryption.
- A5 family (A5/0, A5/1, A5/2, A5/3): Encrypts voice and data over the air interface between MS and BTS.

Contribution to Security and Privacy:

- Subscriber authentication (A3): Prevents unauthorized access to the network.
- Session key generation (A8): Ensures that encryption keys are unique per session, preventing eavesdropping.
- Air interface encryption (A5): Protects voice and data confidentiality, making intercepted signals unreadable.

- **Describe the functions of the MS and SIM. Why does GSM separate the MS and SIM? How and where is user-related data represented/stored in the GSM system? How is user data protected from unauthorized access, especially over the air interface? How could the position of an MS (not only the current BTS) be localized? Think of the MS reports regarding signal quality.**

Describe the functions of the MS and SIM. Why does GSM separate them?

- MS (Mobile Station): Handles radio transmission/reception, signal processing, and user interface (voice, SMS, data).
- SIM (Subscriber Identity Module): Stores subscriber identity (IMSI), authentication keys (Ki), and security info.
- Reason for separation: Allows user portability—the SIM can be moved to another device while keeping the subscriber profile intact.

How and where is user-related data represented/stored in the GSM system?

- ✓ HLR (Home Location Register): Permanent subscriber info and service subscriptions.
- ✓ VLR (Visitor Location Register): Temporary info while roaming.
- ✓ AUC (Authentication Center): Stores encryption keys for secure communication.
- ✓ EIR (Equipment Identity Register): Stores device IMEI for validity.

How is user data protected from unauthorized access, especially over the air interface?

- ✓ Authentication (A3 algorithm): Confirms subscriber identity.
- ✓ Session key generation (A8): Creates unique encryption keys per session.
- ✓ Air interface encryption (A5): Encrypts voice and data between MS and BTS.
- ✓ SIM security: Stores sensitive keys securely; not accessible by the MS directly.

How could the position of an MS (not only the current BTS) be localized?

- ✓ Cell ID: Basic location via serving BTS.
- ✓ Timing Advance (TA): Estimates distance from serving BTS.
- ✓ Neighbor cell measurements: MS reports signal strength/quality from neighboring BTS.
- ✓ Triangulation / network-assisted localization: Uses multiple BTS measurements to improve accuracy.
- ✓ What is protected by the GSM security mechanisms?

- **What is missing compared to today's communication systems?**

End-to-end encryption, strong mutual authentication, and protection against modern attacks like IMSI catchers.

- ❖ **What multiplexing schemes are used in GSM for what purposes? Think also of other layers apart from the physical layer.**

<u>**Physical Layer:**</u>

- ➢ FDM (Frequency Division Multiplexing): divides the spectrum into 200 kHz carrier channels.
- ➢ TDMA (Time Division Multiple Access): each carrier has 8 time slots to serve multiple users.

<u>**Data Link / Network Layers:**</u>

- ➢ Statistical Multiplexing: used for packet-switched services (e.g., GPRS) to share resources efficiently.
- ➢ Logical Channels Multiplexing: multiple logical channels (control, voice, SMS) are mapped onto physical time slots.

- ❖ **Why use a separated, specialized system like TETRA instead of the normal cellular network?**

<u>**Reasons for TETRA:**</u>

- ➢ Very fast call setup for emergency services.
- ➢ Group calls and push-to-talk functionality.
- ➢ High reliability and robustness in harsh environments.
- ➢ Secure communication with encryption.

<u>**Reasons against a separate system:**</u>

- ➢ Requires dedicated infrastructure, increasing cost.
- ➢ Cellular networks can offer voice and data services with some extras.
- ➢ Maintenance of two separate networks can be more complex.

- ❖ **Compare the handovers in GSM with the ones in UMTS. Why are they called hard or soft, respectively?**

- <u>**GSM:**</u>
  - **Hard handover** – "break-before-make." Connection to the current cell is released before connecting to the new cell.
  - Reason: GSM uses **TDMA/FDM**, so a time slot cannot be shared simultaneously between cells.
- <u>**UMTS:**</u>
  - **Soft handover** – "make-before-break." Connection to the new cell is established **before** releasing the old one.
  - Reason: UMTS uses **CDMA**, allowing the mobile to communicate with multiple cells at the same time.

- ❖ **Why did most GSM operators pick UMTS as 3rd generation and cdmaOne operators pick cdma2000?**

<u>**GSM → UMTS:**</u>

- ➢ GSM operators wanted backward compatibility with 2G networks.
- ➢ UMTS (W-CDMA) allowed a smooth migration path, reusing spectrum and core network concepts.

<u>**cdmaOne → cdma2000**</u>:

- ➢ cdmaOne operators preferred evolution within the CDMA family.
- ➢ cdma2000 maintained compatibility with existing CDMA networks and devices.

Reason: Operators chose the 3G standard that best preserved investment and allowed network continuity.

- ❖ **While the data rates increased over time, the delay remained more or less constant at about 100–150 ms from a mobile device to the base station. What problem arises if, e.g., downloading content using TCP via these fast connections?**
  - ➢ TCP uses Round-Trip Time (RTT) for congestion control and flow control.
  - ➢ Even with high data rates, a constant 100–150 ms delay limits the speed at which TCP can acknowledge packets, slowing down throughput.
  - ➢ <u>Problem:</u> TCP cannot fully utilize the high bandwidth, leading to suboptimal download speeds despite fast connections.

❖ Name high-level differences between LTE and GSM/GPRS/UMTS! What do they have in common?

Feature	GSM / GPRS / UMTS	LTE	Common Points
Generation	2G / 2.5G / 3G	4G	Cellular networks
Access	TDMA/FDM (GSM), WCDMA (UMTS)	OFDMA (DL), SC-FDMA (UL)	Use radio spectrum to connect MS
Data Rate	Up to ~2 Mbps	Up to hundreds of Mbps	Voice & data services
Network	CS for voice, PS for data	Fully packet-switched	Base stations & core network
Handover	GSM: Hard, UMTS: Soft	Mostly hard	Mobility management
Latency	~100–150 ms	~10–30 ms	Signaling & scheduling
Services	Voice, SMS, MMS, low-speed data	High-speed data, VoLTE, streaming	Subscriber authentication & security

❖ Compare the architecture of LTE / UMTS / GSM – what is new, what is basically the same?

- New in LTE: Fully packet-switched network, simplified architecture, eNodeB integrates radio and control functions, no separate RNC.
- Same: Core concepts like MS, base stations, mobility management, subscriber authentication remain.
- UMTS vs GSM: UMTS adds WCDMA, RNC for control; GSM uses TDMA/FDM and MSC/VLR for circuit-switched calls.

❖ Compare multiplexing / multiple access in LTE with UMTS / GSM – differences / similarities

Layer	GSM	UMTS	LTE	Similarity
Multiple Access	TDMA/FDM	WCDMA	OFDMA (DL), SC-FDMA (UL)	All divide resources to serve multiple users
Multiplexing	Time slots per carrier	Spreading codes	Subcarriers + time slots	All map multiple logical channels to physical resources
Purpose	Share frequency channels	Share spectrum with codes	Efficient spectrum usage, high data rates	Enable multiple users to share the network

Exercises.

1. Name some key features of GSM, DECT, TETRA, and UMTS. Which features do the systems have in common? Why have the three older systems been specified? In what scenarios could one system replace another? What are the specific advantages of each system?

- **GSM:** Wide area coverage, 9.6–50 kbit/s, supports voice, SMS, MMS.
- **DECT:** Local coverage, supports voice and data, high user density.
- **TETRA:** Regional coverage, ad-hoc mode, very fast connection setup, group calls, voice and data, robust.
- **UMTS:** Medium coverage, higher data rates (384 kbit/s), flexible bandwidth.
- **Common features:** Circuit-switched voice, integrated into fixed networks, ISDN core network.
- **Reasons for older systems:** Optimized for specific scenarios (local, regional, high density).
- **Replacement:** GSM can replace DECT if licensing allows; modified GSM can replace TETRA (except ad-hoc mode).
- **Advantages:** GSM—wide coverage, TETRA—fast setup & ad-hoc, DECT—high density, UMTS—higher data rates.

2. What are the main problems when transmitting data using wireless systems made for voice? How can these problems be mitigated and efficiency improved? How can billing support this?

- **Problems:** Fixed bandwidth for maximum expected rate, long connection setup, delays in spontaneous data transmission.
- **Mitigation:** Introduce packet-switched services (like GPRS).
- **Billing:** Volume-based billing; application-based billing is better since users care about apps, not bytes.

3. Which types of different services does GSM offer? Why are these services separated? Give examples.

- **Bearer services:** Provide communication channels for data/voice.
- **Tele services:** Voice, SMS, MMS.
- **Supplemental services:** Call forwarding, call waiting, call barring.
- **Reason for separation:** Phased introduction, clear roles for network providers, service providers, and device manufacturers.

4. Why is the data rate available to a user lower than the maximum air interface rate in GSM?

- Forward error correction, signaling overhead, guard spaces, limited number of channels.

5. Name the main elements of GSM system architecture and describe their functions. What are the advantages of specifying internal interfaces?

- **MS (Mobile Station):** Device functions—radio, coder/decoder, ID.
- **BSS (Base Station Subsystem):** Connects MS to network, handles radio.
- **NSS (Network & Switching Subsystem):** Core network—switching, HLR, VLR.
- **Advantage of internal interfaces:** Multi-vendor support, flexibility, equipment from different vendors can interoperate.

6. Describe the functions of the MS and SIM. Why does GSM separate MS and SIM? Where is user-related data stored? How is it protected? How can the MS location be determined?

- **MS:** Device-related functions (radio, coders).
- **SIM:** Subscriber info, PIN, authentication.
- **Separation:** Users can change phones while keeping personal data.
- **Data storage:** SIM, VLR (temporary location), HLR (home network).
- **Protection:** Encryption over air, PIN protection, AuC-secured authentication.
- **Localization:** Signal strength from surrounding base stations, timing info.

7. How does the HLR/VLR architecture limit scalability in terms of users, especially moving users?

- VLR can store ~1 million users.
- Frequent movement causes high update load on HLR, reducing scalability.

8. Why is new infrastructure needed for GPRS but not for HSCSD? Which new components are introduced and their purpose?

- **HSCSD:** Uses existing GSM core.
- **GPRS:** Requires SGSN & GGSN routers (IP routing), user context, subscriber registry.

9. What are the limitations of a GSM cell in terms of diameter and capacity (voice/data) for GSM, HSCSD, and GPRS? How can capacity be increased?

A **GSM cell** typically has a diameter of **1–35 km** (urban cells ~1–5 km, rural cells up to 35 km). Its capacity depends on the technology used: **GSM** supports **8–16 simultaneous voice calls per cell** using TDMA; **HSCSD** increases data rates by combining multiple time slots (up to ~57.6 kbps); **GPRS** allows packet-switched data with speeds up to **40–115 kbps** per user.

Capacity can be increased by:

- **Cell splitting** (creating smaller cells)
- **Sectorization** (dividing a cell into sectors with separate antennas)
- **Adding more frequency channels**
- **Using higher-order technologies** like EDGE or LTE for higher data rates.

10. What multiplexing schemes are used in GSM and their purposes?

- **SDM:** Cell layout, frequency reuse.
- **FDM:** Assign channels/frequencies to operators and cells.
- **TDM:** Assign time-slots to terminals for transmission.

11. How is synchronization achieved in GSM and why is it important?

In **GSM**, synchronization is achieved using **timing advance**, **frequency correction**, and signals from the **Broadcast Control Channel (BCCH)**. The **Mobile Station (MS)** synchronizes its transmission with the **Base Transceiver Station (BTS)** to ensure time-aligned TDMA slots.

Importance: Proper synchronization prevents **slot overlap, interference, and call/data errors**, ensuring efficient and reliable communication.

12. What are the reasons for delays in GSM for packet data traffic? Compare circuit-switched and packet-switched.

- **Circuit-switched:** Connection setup (seconds), FEC (~100 ms), propagation.
- **Packet-switched:** Channel access (depends on load), FEC, routing delays.

13. Where and when can collisions occur in GSM? Compare GSM, HSCSD, GPRS.

- Collision occur: During connection setup (slotted Aloha).
- After slot allocation, no collisions occur in data transmission for GSM, HSCSD, or GPRS.

14. Why and when are different signaling channels needed in GSM?

- **SDCCH:** Setup & authentication before TCH.
- **SACCH:** Monitor signal quality during TCH.
- **FACCH:** Extra signaling using TCH slots during ongoing calls.

15. How is localization, location update, and roaming handled in GSM?

In **GSM**, **localization** finds the mobile's approximate location using the **cell ID**. **Location update** happens when the mobile moves to a new area, informing the **VLR** and **HLR** so calls and messages can reach it. **Roaming** lets the mobile use other networks; the visited network's **VLR** stores temporary data, while the home **HLR** keeps permanent info, ensuring seamless service.

16. Why are so many identifiers/addresses needed in GSM? Distinguish user vs system identifiers.

GSM uses multiple **identifiers/addresses** to manage subscribers, ensure security, and enable efficient routing of calls and messages across networks. Different identifiers serve different purposes:

- **User identifiers**: Identify the subscriber or device itself. Examples include:
  - **IMSI (International Mobile Subscriber Identity):** Unique permanent ID of the subscriber.
  - **MSISDN (Mobile Station ISDN Number):** The phone number used to make/receive calls.
- **System identifiers**: Identify network elements, locations, or temporary connections. Examples include:
  - **TMSI (Temporary Mobile Subscriber Identity):** Temporary ID assigned by the network to protect privacy over the air.
  - **IMEI (International Mobile Equipment Identity):** Identifies the mobile device hardware.
  - **LAI (Location Area Identity):** Identifies the location area for routing and paging.

17. Reasons for handover in GSM and associated problems. Steps and resources for HSCSD/GPRS.

Reasons for Handover in GSM:

- MS moving out of a cell's coverage (**mobility**)
- **Signal quality degradation** or interference
- **Network load balancing** (to avoid congestion)
- **Better service quality** (switching to a stronger BTS)

Problems associated with handover:

- **Call drops** if handover fails
- **Interference** during transition
- **Delay** in data transfer or signaling overhead

Steps for handover:

1. MS measures signal strength and quality of current and neighboring cells.
2. MS reports measurements to the BSC.
3. BSC/MSC decides if handover is needed and selects target cell.
4. Resources in the target cell are allocated.
5. Handover command is sent to MS.

Resources for data transmission:

- **HSCSD:** Allocate additional **time slots** for higher data rate.
- **GPRS:** Allocate **packet-switched channels**, update **PDP context**, and maintain **QoS parameters** for data sessions

18. Functions of authentication and encryption in GSM. How is security maintained?

In GSM, security is ensured through **authentication and encryption**. **Authentication** verifies that the mobile user is legitimate using a secret key in the SIM and the Authentication Center (AuC). **Encryption** protects voice and data during transmission so intercepted signals cannot be read, using algorithms like A5/1 and A5/2. Security is further strengthened by **temporary identities (TMSI)** and frequent key updates, keeping the network safe from unauthorized access.

19. How can higher data rates be achieved in GSM? Differences for HSCSD, GPRS, EDGE?

- **HSCSD:** Combines multiple slots.
- **GPRS:** Packet switching, multiple slots, different coding.
- **EDGE:** New modulation scheme.
- Complexity increases; handover delays and coding delays remain.

20. What limits real device data rates for HSCSD/GPRS compared to theoretical GSM?

- Cannot use all 8 slots simultaneously.
- Switching delay from send/receive mode.
- Not all coding schemes supported.

21. Using 115.2 kbit/s GPRS with 0.5 s delay, how many bytes are in transit? How would TCP behave for a 10 kB transfer?

- 115.2 kbit/s × 0.5 × 2 = 14.4 kB in transit.
- TCP performs poorly for 10 kB (inefficient windowing, high latency).
- Needs adjustments like larger initial sending window.

22. How much of GSM network is used by GPRS? Which elements handle data transfer?

- Uses GSM core for localization and authentication.
- Data transfer done by SGSN and GGSN; MSC not needed.

23. What are typical DECT data rates? How are they achieved? Compare with GSM.

- Data rates: 120 channels, 32 kbit/s each.
- Achieved by: TDMA (24 slots/frame), FDM (multiple channels), SDM (base station placement).
- Comparison: Simpler than GSM, fewer handovers, simpler accounting.

24. Who uses trunked radio systems? Why are they attractive and cheaper than GSM?

- Users of Trunked Radio Systems: Police, fire, ambulance, disaster teams, taxis.
- Why they attractive: Fast setup, group calls, ad-hoc, robust, reliable.
- Cheaper due to fewer base stations, less complex billing/accounting.

25. Summarize main features of 3G/UMTS. How are higher capacities and rates achieved?

- Higher data rates, better voice codecs, CDMA, 2 GHz operation.
- Asymmetrical rates via spreading factors.
- More capacity via powerful modulation, compression, and CDMA.

26. Compare mobile networks in Europe, Japan, China, North America.

Mobile networks vary by region in standards, frequencies, and adoption. **Europe** relies on GSM, UMTS, LTE, and 5G, mainly using 900/1800 MHz and 2100 MHz. **Japan** started with proprietary 2G, then W-CDMA, LTE, and 5G, emphasizing high-speed mobile internet. **China** used GSM, its own 3G TD-SCDMA, and rapidly expanded LTE and 5G on bands like 900/1800/2100 MHz and 2.6 GHz. **North America** has a mix of CDMA and GSM, with LTE and 5G on 850/1900 MHz and 700/1700/2100 MHz. Each region reflects its technology focus: Europe on GSM and roaming, Japan on speed, China on rapid 4G/5G growth, and North America on mixed standards with strong LTE/5G.

27. What disadvantage does OVSF have for flexible rates? How does UMTS offer different rates (FDD/TDD)?

- OVSF: only fixed multiples of 15 kbit/s.
- **FDD:** Adjust spreading factor.
- **TDD:** Request slots for uplink/downlink separately.

28. How are different DPDCH channels distinguished in UTRA FDD and TDD?

- **FDD:** Same spreading codes; each UE has unique scrambling code.
- **TDD:** Cell-specific scrambling codes.

29. How are data streams combined/split in CDMA networks? What about soft handover?

- Done at NodeB or RNC, not MSC.
- Iur interface transfers data between RNCs.
- Soft handover possible using rake receivers and multipath combining.

30. Explain near-far effect in UTRA-FDD and GSM. How is it mitigated?

- **Near-far effect:** Stronger signals from nearby users overpower weaker ones.
- **UTRA-FDD:** Power control 1500×/sec ensures equal signal at base.
- **GSM:** TDMA/FDM separates users, avoiding near-far effect.

Chapter-02 (Rapparot books)

❖ **If a total of 33 MHz of bandwidth is allocated to a particular TDD cellular telephone system which uses two 25 kHz simplex channels to provide half duplex voice and control channels, compute the number of channels available per cell if a system uses (a) 4-cell reuse, (b) 7-cell reuse (c) 12-cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems.**

Given that,

- ✓ Total available bandwidth = 33 MHz
- ✓ Simplex channel bandwidth = 25 kHz
- ✓ Spectrum reserved for control channel = 1 MHz

Each duplex channel requires 2 simplex channels, so the bandwidth per duplex channel = 2 × 25 kHz = 50 kHz per duplex channel.

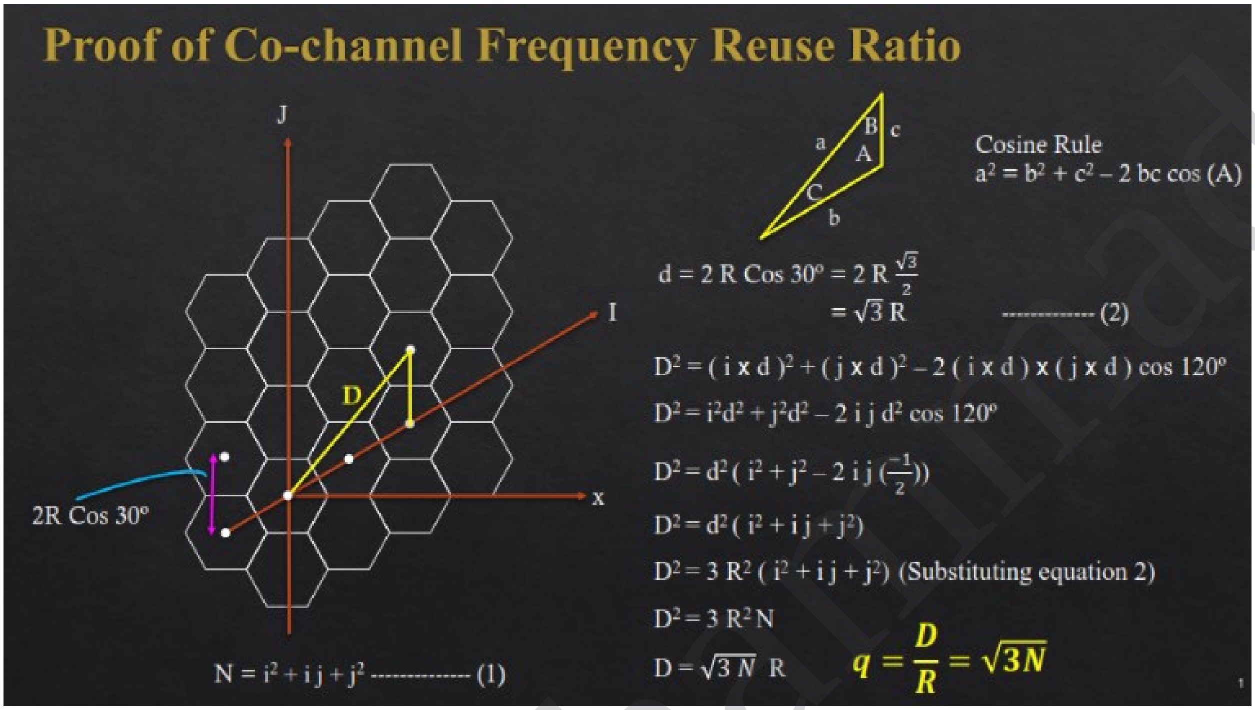
- ✓ Available bandwidth for duplex voice channels = 33 - 1 = 32 MHz
- ✓ Total number of duplex voice channels available = 32,000 kHz / 50 kHz = 640
- ✓ Total number of control half-duplex control channels available = 1000 kHz / 25 kHz = 40

a) For $N = 4$:
Voice Channels per cell = 640/4 = 160
Control channels per cell = 40/4 = 10

b) For $N = 7$:
Voice Channels per cell = 640/7 = 91
Control channels per cell = 40/7 = 6

c) For $N = 12$:
Voice Channels per cell = 640/12 = 53
Control channels per cell = 40/12 = 4

- **Prove that for a hexagonal geometry, the co-channel reuse ratio is given by $Q=\sqrt{3N}$, where $N=i^2+ij+j^2$**



- **In a cellular system with a total of S duplex channels that are divided among N cells, with each cell receiving a unique and disjoint group of k channels, how is the total number of available radio channels expressed? Discuss the impact of cluster size N on co-channel interference and the frequency reuse distance. Additionally, explain how to determine the nearest co-channel neighbors in a hexagonal cell layout with an example.**

If a total of **S duplex channels** are divided among **N cells**, each cell gets

$$k = \frac{S}{N}$$

unique channels.

So, **total channels = S = N × k.**

Effect of Cluster Size (N)

- **Larger N** → Greater distance between co-channel cells → **less interference** and **better signal quality**.
- But each cell gets fewer channels (**lower capacity**).
- **Reuse distance:**

$$D = R\sqrt{3N}$$

where R = cell radius.

Effect of Cluster Size (N)

- **Larger N** → Greater distance between co-channel cells → **less interference** and **better signal quality**.
- But each cell gets fewer channels (**lower capacity**).
- **Reuse distance:**

$$D = R\sqrt{3N}$$

where R = cell radius.

Co-channel Interference

- Increases when N is small (cells are closer).
- Decreases when N is large (cells are farther apart).

$$\mathrm{C/I} \propto (\sqrt{3N})^{\alpha}$$

where α = path loss exponent.

Finding Nearest Co-channel Neighbors

- Cluster size $N = i^2 + ij + j^2$.
- Move **i cells** along one hex axis and **j cells** along another 60° axis to reach the next co-channel cell.
- There are **6 nearest co-channel neighbors**, all at distance D.

❖ **Answer the following:**

i. **What are some of the key performance indicators for 6G networks, and what technologies will enable these targets?**

Key Performance Indicators (KPIs) for 6G and enabling technologies:

- **KPIs:** Data rate > 1 Tbps, latency < 0.1 ms, 99.9999% reliability, massive connectivity, energy efficiency, global coverage, and AI-driven networking.
- **Enabling technologies:** Terahertz (THz) communication, AI/ML for intelligent networks, reconfigurable intelligent surfaces (RIS), quantum communication, edge computing, and advanced MIMO systems.

ii. **What are the key challenges associated with the deployment of mmWave technologies in 5G and 6G networks?**

- **High path loss** and limited coverage range.
- **Poor penetration** through buildings and obstacles.
- **Signal blockage** by weather or human bodies.
- **Complex beamforming** and alignment needed.
- **High infrastructure cost** due to dense small-cell deployment.

❖ **How is the spectrum allocated for 5G deployment, and what are the characteristics of each frequency band used in this allocation?**

The **5G spectrum** is divided into **three main frequency bands**, each with different characteristics:

1. **Low-band (< 1 GHz):**
   - Example: 600 MHz, 700 MHz.
   - **Wide coverage**, good building penetration, but **lower data rates** (up to a few hundred Mbps).
2. **Mid-band (1–6 GHz):**
   - Example: 3.5 GHz, 4.9 GHz.
   - **Balanced** between coverage and capacity.
   - Provides **faster speeds** (1–2 Gbps) with moderate range.
3. **High-band (mmWave, > 24 GHz):**
   - Example: 26 GHz, 28 GHz, 39 GHz.
   - **Extremely high data rates** (up to 10+ Gbps) and **very low latency**, but **short range** and poor obstacle penetration.

❖ **Give reasons for a handover in GSM and the problems associated with it. What are the typical steps for handover, what types of handover can occur? Which resources need to be allocated during handover for data transmission using HSCSD or GPRS respectively? What about QoS guarantees?**

Reasons for handover in GSM:

- When a mobile station (MS) moves out of the current cell's coverage area.
- Signal strength or quality becomes too low.
- To balance traffic load between cells.

Problems with handover:

- Call drops if handover fails or is delayed.
- Signaling delay and resource unavailability.
- Interference and timing issues between cells.

Typical handover steps:

1. MS measures signal levels from neighboring cells.
2. MS sends measurement reports to the serving BTS.
3. BSC decides if handover is needed.
4. Target BTS is allocated and link is prepared.

Types of handover:

1. **Intra-cell:** within same cell (rare).
2. **Intra-BSC (intra-MSC):** between BTSs under same BSC.
3. **Inter-BSC (intra-MSC):** between different BSCs under one MSC.

Resources allocated during handover:

- **For HSCSD (High-Speed Circuit Switched Data):** multiple traffic channels must be reserved in the target cell for continuous data flow.
- **For GPRS (Packet Switched):** temporary block flows (TBFs) and packet data channels are re-established in the new cel

❖ **Comparison between infrastructure vs. ad-hoc vs. mesh networks**

Feature	Infrastructure Network	Ad-hoc Network	Mesh Network
Definition	Uses fixed infrastructure like base stations or access points to connect devices.	Devices communicate directly without any fixed infrastructure.	A hybrid of ad-hoc and infrastructure; devices (nodes) relay data for others, forming a multi-hop network.
Topology	Centralized	Decentralized	Partially decentralized, multi-hop
Setup	Requires pre-deployed infrastructure	Quick and easy to set up	Moderate setup; nodes auto-configure connections
Range	Depends on base station/AP coverage	Limited to node-to-node communication range	Extended via multi-hop links
Reliability	High, due to stable infrastructure	Lower, due to mobility and limited range	High, as multiple paths increase redundancy
Example	Cellular networks, Wi-Fi	Bluetooth, MANET	Wireless community networks, IoT mesh systems